JEE Mains 2019 Chapter wise Question Bank

Properties of Triangle – Questions

Q1

With the usual notation, in ΔABC, if $\angle A + \angle B = 120°$, $a = \sqrt{3}+1$ and $b = \sqrt{3}-1$, then the ratio $\angle A : \angle B$, is:

(1) $7:1$ (2) $5:3$

(3) $9:7$ (4) $3:1$

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Q2

In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y. if $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is :

(1) $\frac{3}{2}y$ (2) $\frac{c}{\sqrt{3}}$ (3) $\frac{c}{3}$ (4) $\frac{y}{\sqrt{3}}$

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Q3

Given $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ for a ΔABC with usual notation. If $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos C}{\gamma}$, then the ordered triad (α, β, γ) has a value :

(1) $(7, 19, 25)$ (2) $(3, 4, 5)$

(3) $(5, 12, 13)$ (4) $(19, 7, 25)$

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Q4

If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is :

(1) $5:9:13$ (2) $4:5:6$

(3) $3:4:5$ (4) $5:6:7$

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Q5

The angles A, B and C of a triangle ABC are in A.P. and $a : b = 1 : \sqrt{3}$. If $c = 4$ cm, then the area (in sq.cm) of this triangle is :

(1) $\frac{2}{\sqrt{3}}$ (2) $4\sqrt{3}$ (3) $2\sqrt{3}$ (4) $\frac{4}{\sqrt{3}}$

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Properties of Triangle - Solutions

Q1

(1) $\angle A + \angle B = 120°$ ...(1)

$\Rightarrow \angle C = 180° - 120° = 60°$

$$\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\left(\frac{C}{2}\right)$$

$$= \frac{2}{2\sqrt{3}}(\cot 30°) = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

$\Rightarrow \frac{\angle A - \angle B}{2} = \frac{\pi}{4}$ ($\angle$ is angle)

$\Rightarrow \angle A - \angle B = 90°$ ...(2)

From eqn (1) and (2)

$\angle A = 105°$, $\angle B = 15°$

Then, $\angle A : \angle B = 7 : 1$

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Q2

(2) Let two sides of triangle are a and b.

$$a + b = x$$

$$ab = y$$

$x^2 - c^2 = y \Rightarrow (a+b)^2 - c^2 = ab$

$\Rightarrow (a+b-c)(a+b+c) = ab$

$\Rightarrow 2(s-c)(2s) = ab$

$\Rightarrow 4s(s-c) = ab$

$\Rightarrow \frac{s(s-c)}{ab} = \frac{1}{4}$

$\Rightarrow \cos^2\frac{c}{2} = \frac{1}{4}$

$\Rightarrow \cos c = -\frac{1}{2} \Rightarrow c = 120°$

$\therefore$ Area of triangle is,

$$\Delta = \frac{1}{2}ab(\sin 120°) = \frac{\sqrt{3}}{4}ab$$

$\because \quad R = \frac{abc}{4\Delta}$

$\therefore \quad R = \frac{abc}{\sqrt{3}\,ab} = \frac{c}{\sqrt{3}}$

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Q3

(1) Let $\dfrac{b+c}{11} = \dfrac{c+a}{12} = \dfrac{a+b}{13} = k$ (Say).

$\therefore \quad b + c = 11k,\ c + a = 12k,\ a + b = 13k$

$\therefore \quad a + b + c = 18k$

$\therefore \quad a = 7k,\ b = 6k$ and $c = 5k$

$$\therefore \quad \cos A = \frac{36k^2 + 25k^2 - 49k^2}{2.30k^2} = \frac{1}{5}$$

$$\text{and } \cos B = \frac{49k^2 + 25k^2 - 36k^2}{2.35k^2} = \frac{19}{35}$$

$$\text{and } \cos C = \frac{49k^2 + 36k^2 - 25k^2}{2.42k^2} = \frac{5}{7}$$

$\therefore \quad \cos A : \cos B : \cos C = 7 : 19 : 25$

$$\therefore \quad \frac{\cos A}{7} = \frac{\cos B}{19} = \frac{\cos C}{25}$$

Hence, required ordered triplet is (7, 19, 25).

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Q4

(2) Let the sides of triangle are $a > b > c$ where
Given $A = 2C$
$\because A + B + C = \pi$ and $A = 2C$
$\Rightarrow B = \pi - 3C$ ...(i)
$\because a, b, c$ are in A.P. $\Rightarrow a + c = 2b$
$\Rightarrow \sin A + \sin C = 2 \sin B$ ...(ii)
$\Rightarrow \sin A = \sin (2C)$ and $\sin B = \sin 3C$
From (ii),
$\sin 2C + \sin C = 2 \sin 3C$
$\Rightarrow (2\cos C + 1) \sin C = 2 \sin C\, (3 - 4 \sin^2 C)$
$\Rightarrow 2\cos C + 1 = 6 - 8 (1 - \cos^2 C)$
$\Rightarrow 8\cos^2 C - 2\cos C - 3 = 0$

$\Rightarrow \cos C = \dfrac{3}{4}$ or $\cos C = -\dfrac{1}{2}$

$\because$ C is acute angle

$\Rightarrow \cos C = \dfrac{3}{4} \quad \Rightarrow \sin C = \dfrac{\sqrt{7}}{4}$

and $\sin A = 2 \sin C \cos C = 2 \times \dfrac{\sqrt{7}}{4} \times \dfrac{3}{4} = \dfrac{3\sqrt{7}}{8}$

$\sin B = \dfrac{3\sqrt{7}}{4} - \dfrac{4\sqrt{7}}{4} \times \dfrac{7}{16} = \dfrac{5\sqrt{7}}{16}$

$\Rightarrow \sin A : \sin B : \sin C :: a : b : c$ is $6 : 5 : 4$

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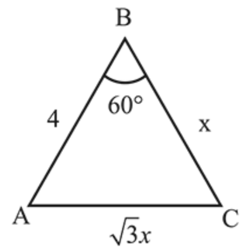
Q5

(3) Given that A, B, C, are in A.P. $\Rightarrow 2B = A + C$

Now, $A + B + C = \pi \Rightarrow B = -$

Area $= \dfrac{1}{2}(4x)\sin 60° = \sqrt{3}x$



Now $\cos 60° = \dfrac{16 + x^2 - 3x^2}{8x}$

$\Rightarrow 4x = 16 - 2x^2 \Rightarrow x^2 + 2x - 8 = 0$
$\Rightarrow x = 2$ [$\because x$ can't be negative]
Hence, area $= 2\sqrt{3}$ sq. cm

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